

JOSÉ ALFREDO ROSAS CÓRDOVA

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EDUCATION

Bachelor of Aeronautical Engineering <i>Universidad Autónoma de Nuevo León</i> Graduated summa cum laude GPA: 97.97	08/2017 — 06/2023
Combined Study and Practice Stays (DAAD-KOSPIE) Bachelor of Mechanical Engineering <i>Technische Universität Braunschweig</i> GPA: 1.9	09/2021 — 09/2022

LANGUAGES

Spanish	Mother tongue
English	TOEFL ITP – 677 Points (C1)
German	TestDaF – TDN 5 (C1)

IT SKILLS

CAD & CAM:	PTC Creo, OnShape, SOLIDWORKS, Fusion 360
Programming:	Python – Experienced MATLAB – Experienced
Simulation & Modelling:	ANSYS Workbench, Fluent, Simulink, Xfoil
Others:	MS Office, Latex, GitHub, Oracle PLM & ERP

WORK EXPERIENCE

Mechanical Engineer <i>New Product Development & Introduction, Liquid Cooling</i> <i>Vertiv – Monterrey, Mexico</i> • Development, documentation and sustaining mechanical components and assemblies such as sheet metal welded structures; primary and secondary cooling circuits (i.e. refrigerant and water copper and stainless steel based piping networks, respectively) for new products tailored to high density data center cooling applications (e.g. chillers, CDUs, fluid networks). All within a collaborative work environment based on the use of PLM and ERP systems based on the use of Engineering Change Orders and strict revision control. • Execution of both DFM& A as well as DFMEA during the design and development process for the identification of areas of opportunity before production release. • Participation in design validation testing for both complete, production-ready units as well as engineering prototypes for both performance assessments and certification. • Collaboration with manufacturing and field teams and facilities for the implementation of design improvements around manufacturing as well as serviceability concerns within both design and production environments. • Participation in the planning, part procurement, and build process of both engineering, and manufacturing prototypes of brand new units both locally and abroad. • Processing of design validation test data for verifying the performance of new products.	06/2023 — today
Project Engineer - UAV Development <i>Spacelab MX – Monterrey, Mexico</i> • Initial aircraft sizing and preliminary flight performance studies. • Design of the initial airframe structural for the prototype phase of the project.	01/2023 — 04/2023

- Integration, ground-, and flight testing of the testbed aircraft.

System Testing & Mechanical Design Intern, Engineering Mechatronics Department
Bosch Rexroth AG – Stuttgart, Germany

04/2022 — 08/2022

- Design and implementation of system tests within a production environment for autonomous mobile robots (UGV) based around previously defined KPIs established around beta-tester customer requirements.
- Evaluation, analysis, and fault detection of the drive performance of the UGVs as well as the communication layer between UGVs and fleet management service based on test data.
- Handling and maintenance of autonomous mobile robots.
- Mechanical design of internal hardware elements and drive mechanisms.
- Contribution to the continuous improvement of product quality.

RESEARCH EXPERIENCE

Research Assistant, Aerodynamics Laboratory

07/2020 — 06/2023

Centro de Investigación e Innovación en Ingeniería Aeronáutica – Monterrey, Mexico

- Scientific and technological research funding program of the UANL
 – *Study of Unmanned Aerial Vehicles*. Project number: IT1814-21 05/2021
- XXII Sommer programm for scientific and technological research
 PROVERICYT 07/2020
 – *Novel control methods for small aircraft*
- General maintenance work of manufacturing equipment. Mostly but not limited to CNC machines (e.g hot-wire cutters, table routers and 3D printers).
- Co-published potential flow solver validation paper on the *Revista Internacional de Métodos Numéricos para Cálculo y Diseño en Ingeniería* journal. 11/2024
 – *Validation of VSPAERO for Basic Wing Simulation* (DOI: 10.23967/j.rimni.2024.10.56492).

EXTRACURRICULAR ACTIVITIES

Chief Engineer, Axios Aero Design

07/2021 — 06/2022

Student Team

- Aircraft conceptual and preliminary design studies encompassing aerodynamics, structures, electric propulsion, and flight stability and control.
- Project planning and execution based around system engineering methodologies.
- Assistance in the detail design of structural aircraft components such as wing, fuselage, and empennage structures.

Technical Report Lead, Axios Aero Design

09/2020 — 06/2021

Student Team

- Technical report production consisting primarily of a documentation of the engineering design process as well as technical drawings describing the final product.

AWARDS

SAE Aero Design México 2022

03/2022

2. Place Overall

3. Place Technical Report

MathWorks Minidrone Competition Mexico 2021

08/2021

1. Place